

Pakistan EV & Battery Industrialization Policy, Market, Supply Chain & FDI Tracker

Evidence snapshot: 28 August 2026

EXECUTIVE VERDICT

Pakistan has moved from an EV-import story toward an assembly-and-localization story, but it has not yet crossed the line into a fully domestic battery manufacturing ecosystem. The decisive gap is cells and upstream materials processing. BYD–Mega Motor is a real, funded CKD assembly project; Treet–CNTE is a real MoU with a CATL-backed technology company. Direct CATL manufacturing in Pakistan remains a discussion-stage proposition, not a confirmed cell gigafactory.

Prepared for investor, policy and market-intelligence use. Figures are time-stamped and confidence-labelled; draft policy claims are explicitly separated from notified rules.

1. Executive dashboard

Indicator	Latest signal	Investor meaning	Confidence
Battery imports	4.6 GWh in 2025; 8.75 GWh projected by 2030	Large near-term demand pool for local packs, BMS, testing and eventually cells	High / source-based
Battery policy	LFP selected; localization roadmap circulated; final framework still evolving	JV terms should be staged against notified incentives	Medium
Auto policy	In-principle consensus reported Aug-2026; not yet fully notified	Do not underwrite long-term tariff treatment until gazetted	Medium
Charging	~30–50 public stations estimated; 3,000 target by 2030	Infrastructure remains a binding constraint outside major cities	High
BYD–MMC	\$150m plant; 25,000 units/yr; H2 2026 operations expected	Confirmed manufacturing/assembly anchor	High
Treet–CNTE	MoU signed May 25, 2026; CATL-backed CNTE	Real entry point for CATL-linked technology, but not cell production	High

THE ONE-LINE INVESTOR TEST

If a project imports cells and only builds packs, classify it as assembly/value addition. If it makes electrochemical cells from active materials and has local process control, classify it as manufacturing. Pakistan currently has far more evidence of the first than the second.

2. Policy & incentive tracking

2.1 National Lithium-Ion Battery Manufacturing Policy 2026–31

Public reporting shows a policy process rather than a fully settled legal regime. A January 2026 working-group process approved LFP (lithium iron phosphate) as the initial chemistry for localization and emphasized testing, certification and recycling. By August 2026, the Ministry was still developing EV-battery standards while the broader Next Generation Energy Storage Policy was being finalized. This means investors should distinguish policy direction from enforceable tariff rights.

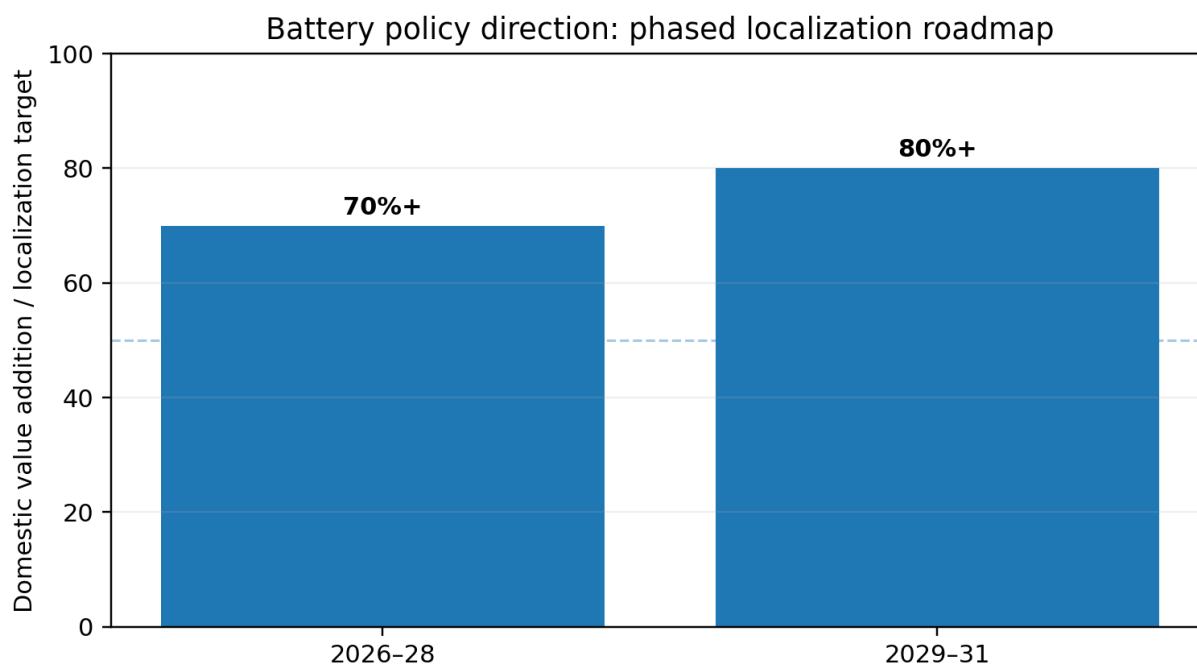


Chart 1. Localization roadmap reported in Pakistan's investor pitchbook. The 70%+/80%+ figures should be treated as policy-direction/investment-pitchbook targets, not as a substitute for a Gazette notification.

Investor interpretation: the most bankable structure is a staged JV. Stage 1 can legitimately use imported cells with local module/pack assembly, BMS integration and testing; Stage 2 should add local electrodes/material processing; Stage 3 should be conditioned on cell-line economics, qualification and recycling. A JV that prices itself as a 'gigafactory' while importing finished cells should not receive the same valuation or policy treatment as a true cell manufacturer.

2.2 Tariff and incentive schedule — what is actually firm

Instrument	Status	Published / reported rate	Applies to	Investor reading
EV CKD: EV-specific parts	Notified framework, legacy extended to Jun-2027	1%	EV CKD specific components	Real but time-bounded; confirm renewal/conditions
EV CKD: non-localized parts	Notified framework	10%	Non-localized CKD	Creates a tariff incentive to localize
EV CKD: localized parts	Notified framework	25%	Localized CKD parts	Shows why 'localization' must be economically meaningful
Battery policy: cells	Policy direction / pitchbook	0% cells reported until 2028	Battery manufacturing inputs	Do not treat as fully enforceable until notified
Battery policy: CBU batteries	Policy direction / pitchbook	20% FED reported	Imported CBU batteries	Designed to encourage local value addition
Battery manufacturing	Policy direction / pitchbook	1% sales tax; up to 90% accelerated depreciation reported	Local manufacturers	Potentially material, but verify final legal text

POLICY RISK FLAG

Investors should not underwrite a project on the assumption that proposed incentives will become legally enforceable. Pakistan's battery-policy framework was still evolving in August 2026. Financial models should therefore use existing, notified concessions as the base case and treat proposed battery-policy incentives as an upside scenario until formally approved and notified.

2.3 EV Battery Policy: confirmed vs draft

Confirmed / evidenced	Still evolving / draft	JV implication
Government is actively finalizing an EV/energy-storage policy	Final legal framework and detailed incentive schedule	Use milestone-based investment tranches
LFP approved for initial localization	Detailed localization tiers and enforcement metrics	Define cell, module and pack KPIs contractually
PSQCA battery standards are being developed	Final EV-battery standards/certification regime	Include certification capex and qualification timeline
Import-dependent battery market is large and growing	Exact treatment of cells, cathodes, anodes and recycling	Keep initial supply chain flexible

2.4 Auto Policy 2026–31 and local content

As of the brief date, public reporting indicates that an Ishaq Dar-led steering committee reached in-principle consensus on the new Auto Policy 2026–31, but the final proposal still had to move through the Ministry of Industries and the federal cabinet. Therefore, a definitive 2026–31 local-content threshold should not be presented as settled unless it appears in the notified policy.

WHAT CAN BE MEASURED TODAY

The expiring AIDEP 2021–26 used a tariff-based localization system rather than a single universal 'EV local content = X%' rule. It also set a 30% local value-addition requirement for parts/components seeking concessions under the relevant framework. Existing vehicle localization examples were much higher for mature models, but high-tech components remained import dependent.

For JV due diligence, define local content in layers: (1) body/chassis, (2) wiring/thermal systems, (3) motor/inverter, (4) BMS, (5) battery module/pack, (6) cells, (7) active materials. A headline 30% local value addition can conceal a project that still imports the most strategic components.

3. Market data

3.1 Is the 61% March EV jump holding?

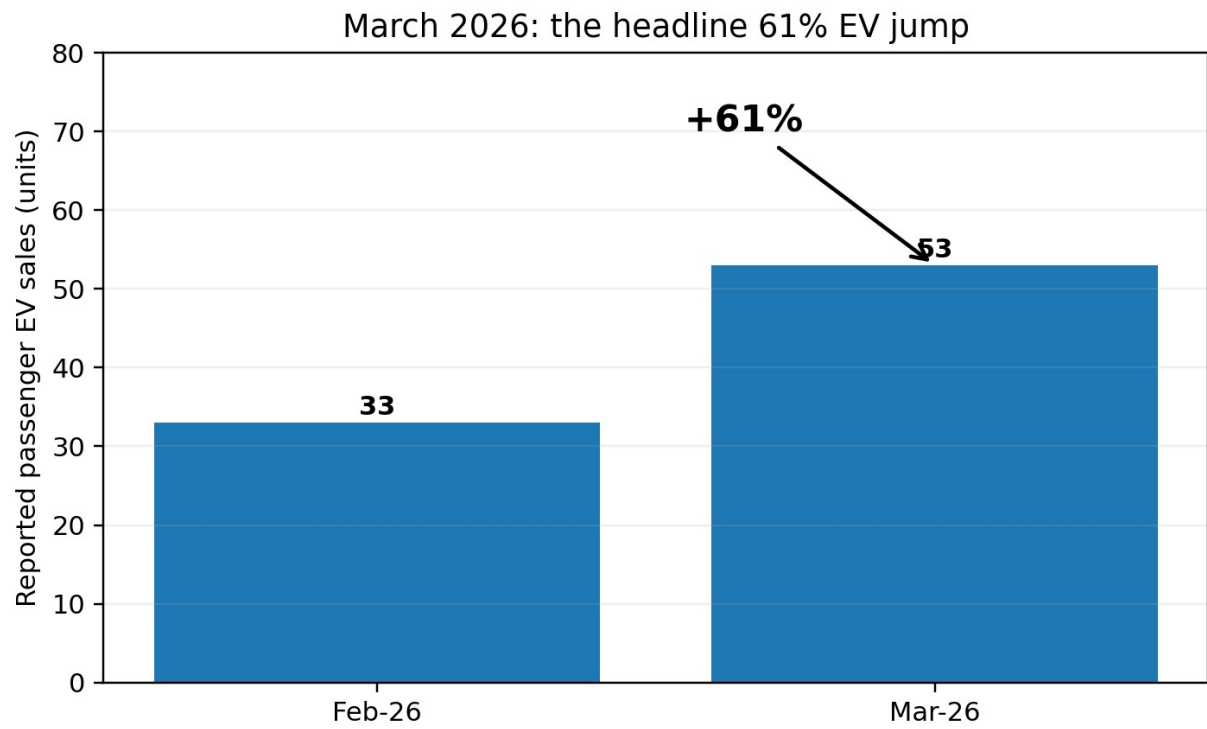


Chart 2. Reported passenger-EV sales rose from 33 units in February to 53 in March 2026 (+61%). The absolute base is small.

The 61% number is real in the cited Arif Habib/Pakistani reporting, but it should not be mistaken for a 61% market share or a national EV-and-hybrid registration series. It is a passenger-EV sales comparison. April data show Honri-Ve sales falling 43% from 53 to 30 units, but that is a model-level observation and cannot prove the entire EV market reversed. The more defensible conclusion is: March confirmed a demand signal, but the public monthly dataset is too fragmented to claim a clean post-March national time series.

DATA GAP — NATIONAL REGISTRATION DATA IS FRAGMENTED

Pakistan does not currently provide a single, easily downloadable national dataset combining monthly EV, HEV and PHEV registrations by brand and province. ARS therefore triangulates market activity using manufacturer sales, provincial registration data, customs/import statistics and company disclosures. This approach reduces the risk of treating a single model or OEM as a proxy for the national market.

3.2 Chinese EV price tiers

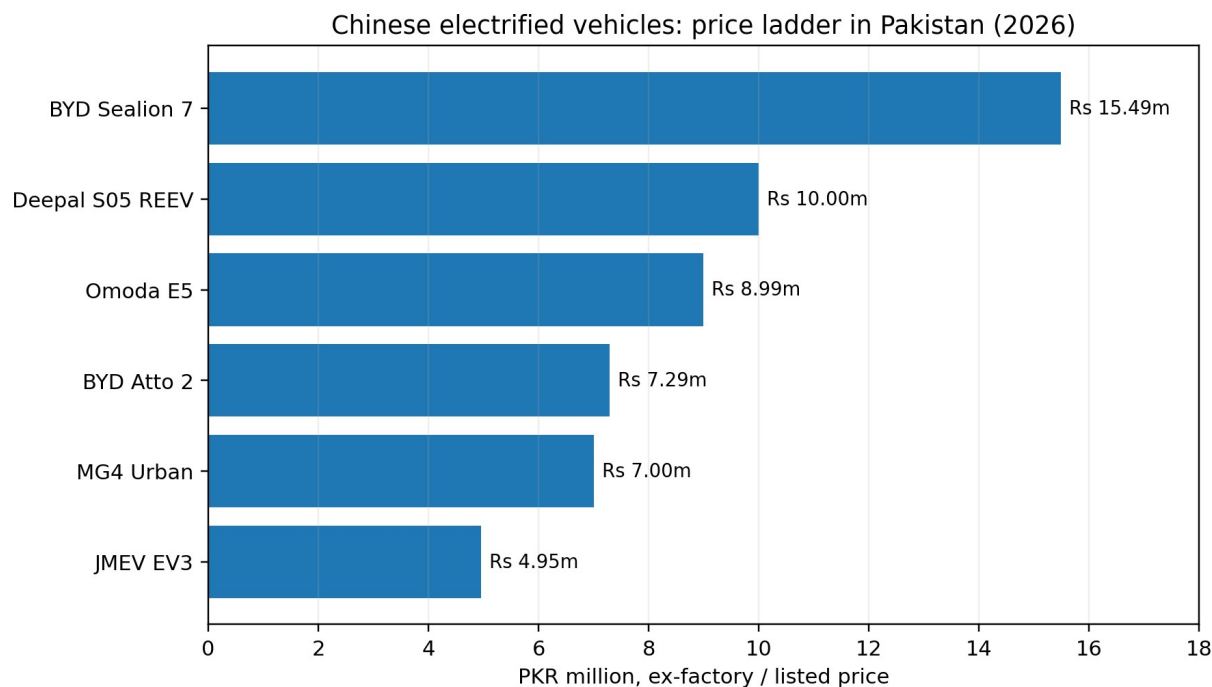


Chart 3. Selected 2026 Chinese electrified-vehicle price ladder in Pakistan. Prices are listed/ex-factory reference points and can change with taxes, offers and model revisions.

The market is no longer one premium EV niche. Entry products are moving toward the Rs5–7 million zone, while mainstream crossovers cluster around Rs7–10 million and larger/premium vehicles move beyond Rs15 million. That widening ladder matters for localization: high-volume affordable models are the best candidates for scale-driven local content, whereas premium CBUs can remain import-heavy.

3.3 Running-cost economics — the chart investors should actually use

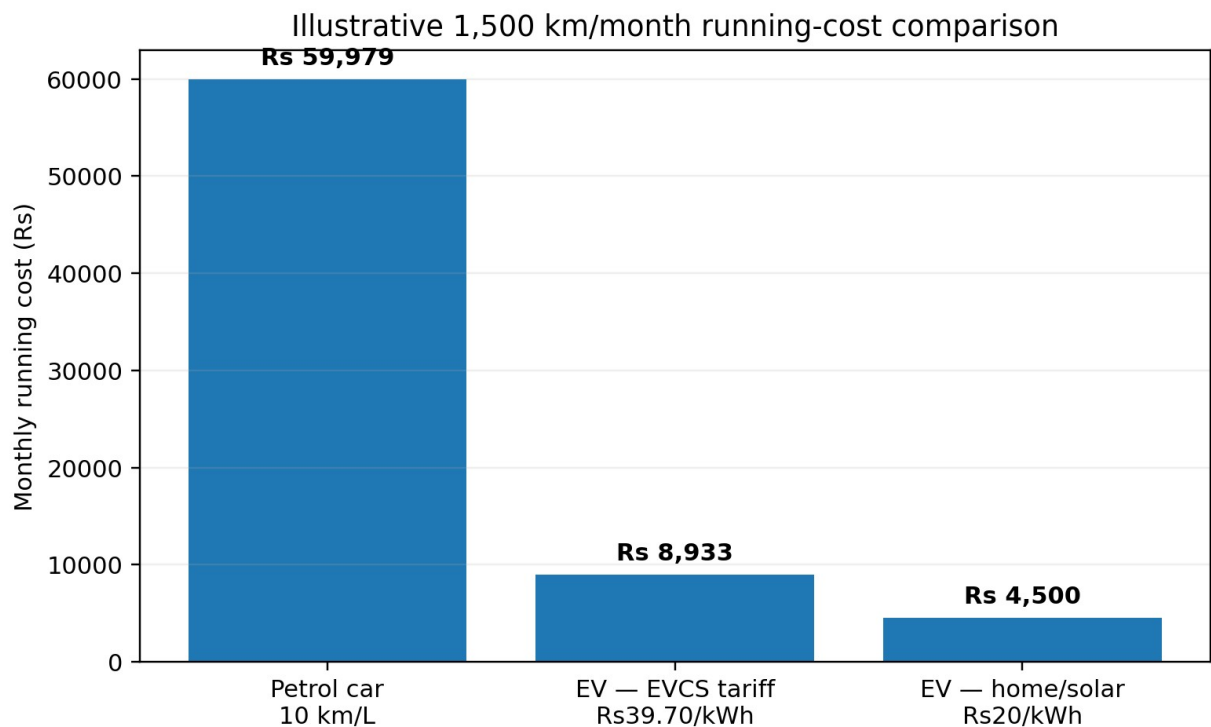


Chart 4. Illustrative 1,500 km/month comparison using petrol at Rs399.86/L, a 10 km/L petrol car, EV consumption of 15 kWh/100 km, and the official EV charging-station tariff of Rs39.70/kWh. The Rs20/kWh case represents a home/solar-style scenario, not the public charging tariff.

At 1,500 km/month, the illustrative petrol bill is about Rs60,000. Charging at the Rs39.70/kWh EVCS tariff would be about Rs8,900, while a Rs20/kWh home/solar assumption produces about Rs4,500. This is why the popular 'Rs4,000 vs Rs65,000' meme is directionally understandable but not universal: it depends heavily on mileage, vehicle efficiency, petrol price and whether charging is done at home, from solar or at a commercial fast charger.

3.4 Charging infrastructure

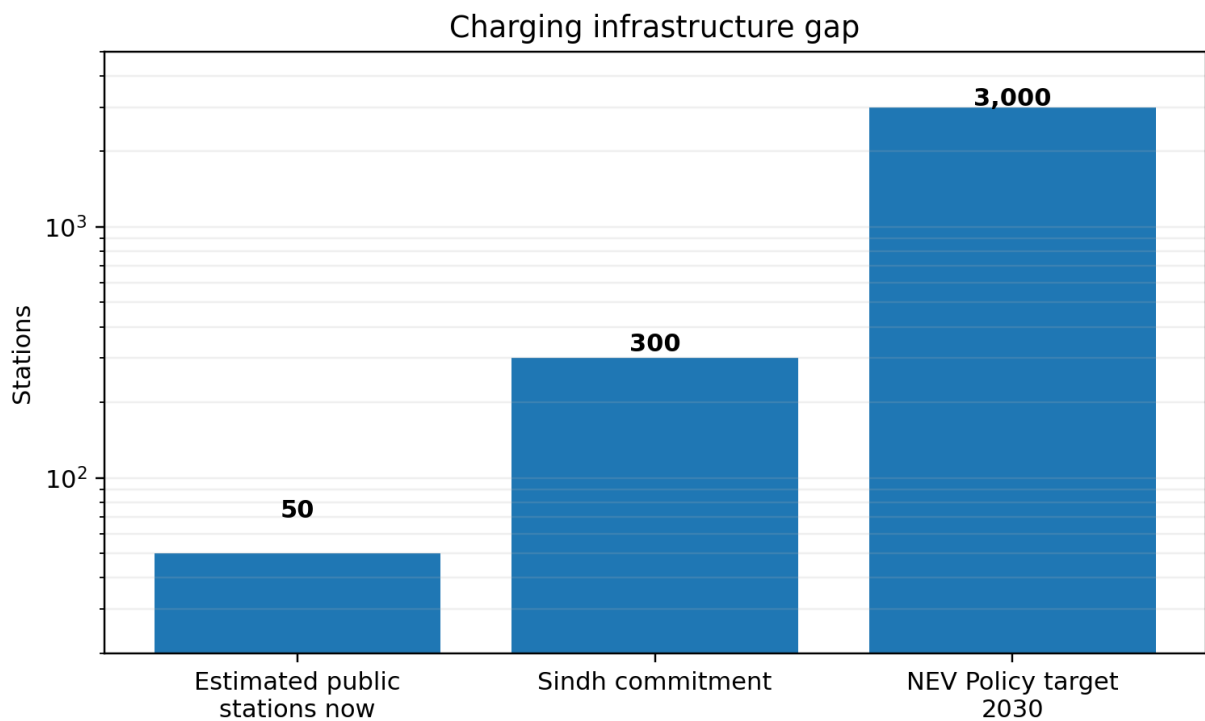


Chart 5. Estimated current public charging base versus Sindh’s reported 300-station commitment and the national 3,000-station 2030 target. Log scale is used because the gap is very large.

Pakistan is estimated to have only around 30–50 public EV charging stations, while the national policy target is 3,000 by 2030. Sindh has separately discussed/committed to a 300-station buildout. Existing commercial infrastructure is concentrated in Islamabad, Lahore and Karachi, with highway coverage improving but still leaving large geographic gaps.

GEOGRAPHIC BOTTLENECK

For an investor, the question is not 'How many chargers exist?' but 'How many chargers exist every 100–150 km on the corridors where customers actually travel?' The next investment wave should target intercity corridors and secondary cities, not only another cluster inside Islamabad/Lahore/Karachi.

4. Supply chain & minerals

4.1 Assembly vs. real manufacturing

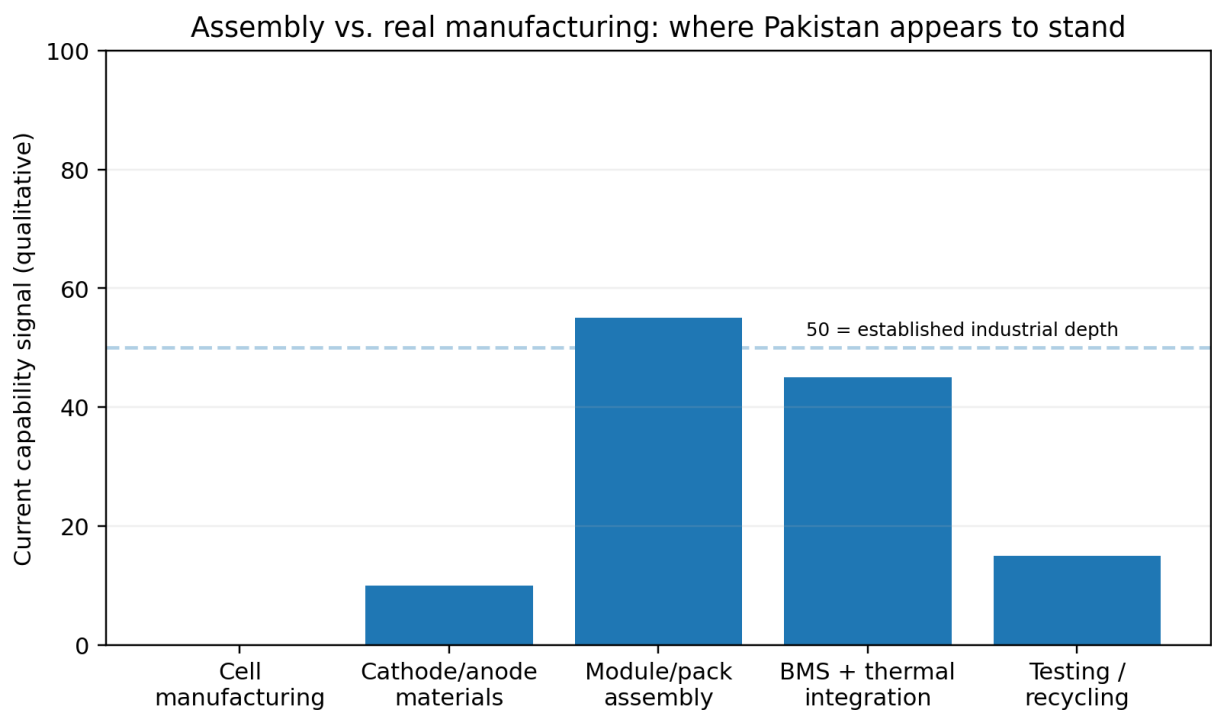


Chart 6. Qualitative capability map. This is an evidence-weighted analytical score, not a measured percentage of national production.

Pakistan's current battery story is best described as an assembly-and-integration market. Public evidence supports pack/system assembly, BMS integration, imported cells and growing storage-system distribution. The strongest warning sign is that public investment material itself identifies cell production, BMS, testing, recycling and raw-material processing as gaps. The Korangi project is important, but its early capacity is only 4 MW and public reporting does not establish a scaled domestic cell industry.

4.2 Cell import dependency

Cell-manufacturing status: import dependence remains dominant

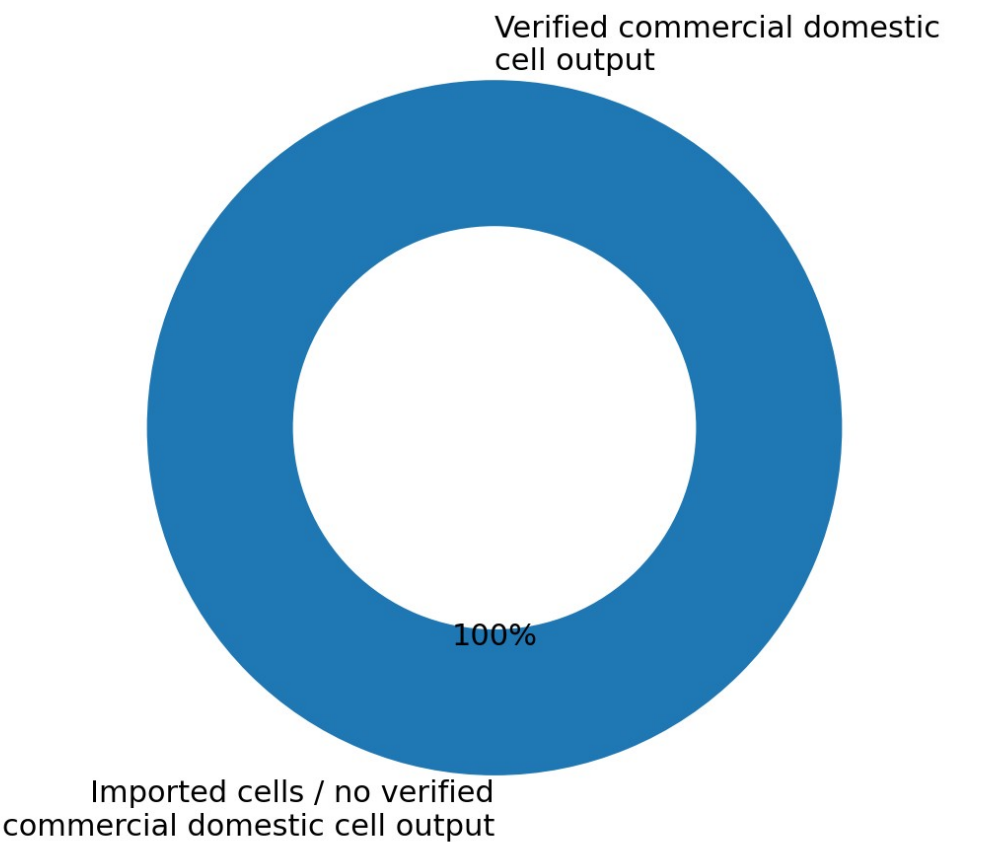


Chart 7. Status view rather than a production-share claim: public investment material reported no local cell manufacturing; current projects are primarily pack/system assembly or planned manufacturing.

The import-dependency story is reinforced by battery-storage imports: Renewables First reported 4.6 GWh of lithium-ion BESS imports in 2025, up 220% year-on-year. Earlier estimates put 2024 lithium-ion battery-pack imports at 1.25 GWh and projected 8.75 GWh by 2030. The implication is that local assembly can scale quickly because the market already exists — but cell manufacturing requires a much deeper process ecosystem.

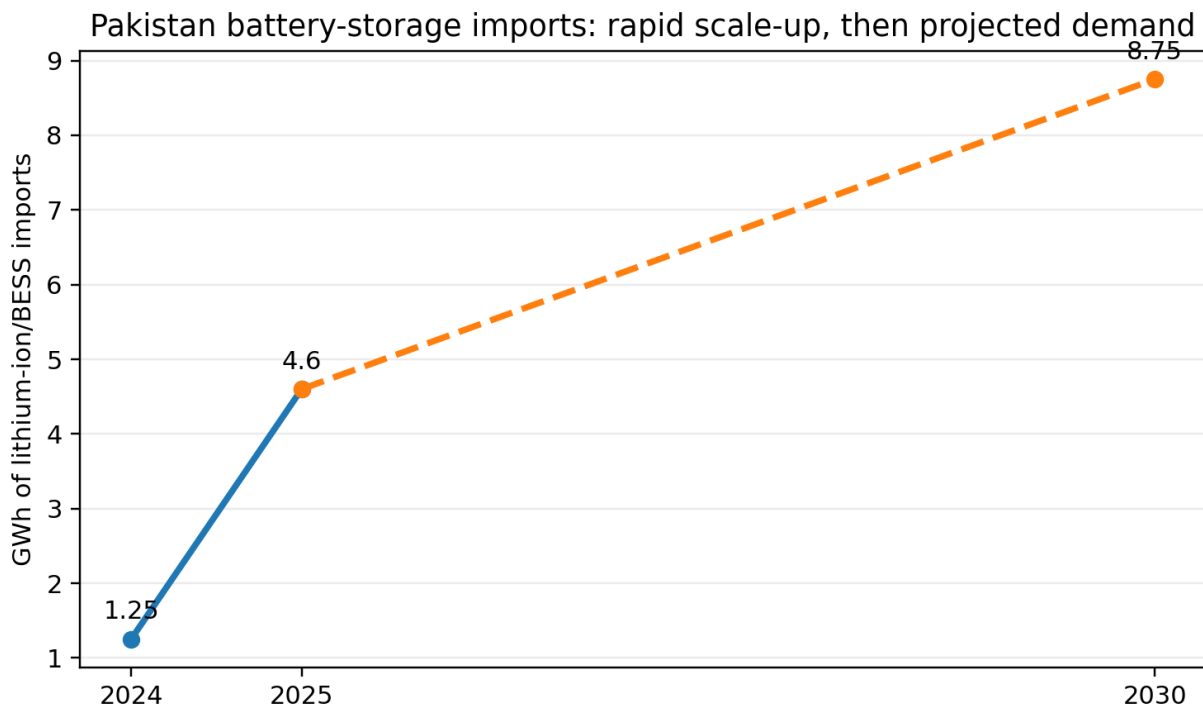


Chart 8. Battery-storage import trajectory: 1.25 GWh in 2024, 4.6 GWh in 2025, and 8.75 GWh projected for 2030.

Import tracking should therefore separate: cells; cathode active material; anode material; electrolyte/separator; modules; finished packs/BESS. A single 'battery imports' line hides where localization is actually occurring.

4.3 Domestic mineral base for LFP

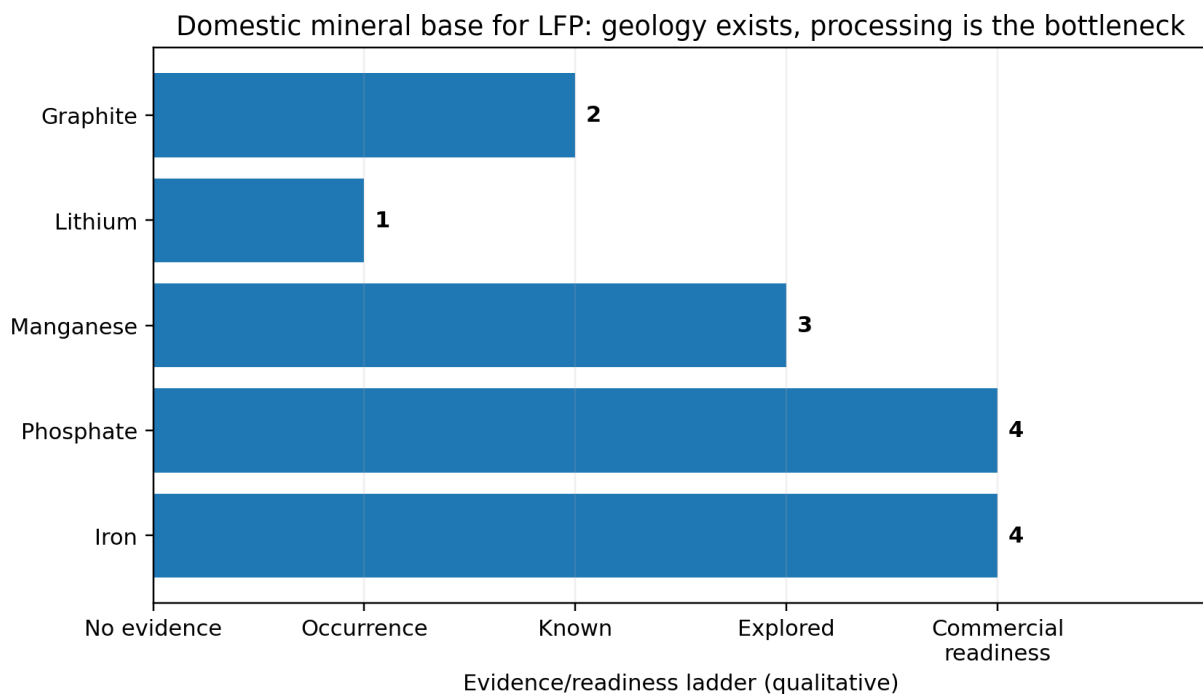


Chart 9. Qualitative readiness assessment. Pakistan has documented mineral occurrences and deposits, but commercial LFP-grade processing is the missing bridge.

Pakistan has a meaningful geological base for several LFP inputs. Iron resources are documented in Punjab, KP and Balochistan; phosphate is mined in the Hazara/Abbottabad district; lithium exploration is ongoing in northern areas and AJK; graphite occurrences are documented in northern Pakistan and KP. The critical distinction is between 'geological occurrence' and 'battery-grade processed material'. For LFP, battery-grade iron/phosphate processing and precursor quality control matter as much as the raw ore itself.

Input	Documented location signal	Battery relevance	Near-term readiness
Iron	Mianwali/Kalabagh, Chiniot, Balochistan	Fe source for LFP	High geological availability; processing upgrade needed
Phosphate	Hazara/Abbottabad; newer Abbottabad occurrences	PO4 source for LFP	Established mining base, but battery-grade processing not yet proven
Manganese	Documented mineral occurrences in Pakistan	Alternative/adjacent cathode chemistries	Resource evidence exists; battery-grade refining gap
Lithium	Northern areas / AJK exploration	Lithium-ion cathode feedstock	Exploration stage; no proven large commercial supply
Graphite	Northern Pakistan/KP occurrences	Anode material	Occurrences known; quality/scale constraints

5. Investor-specific Chinese JV tracker

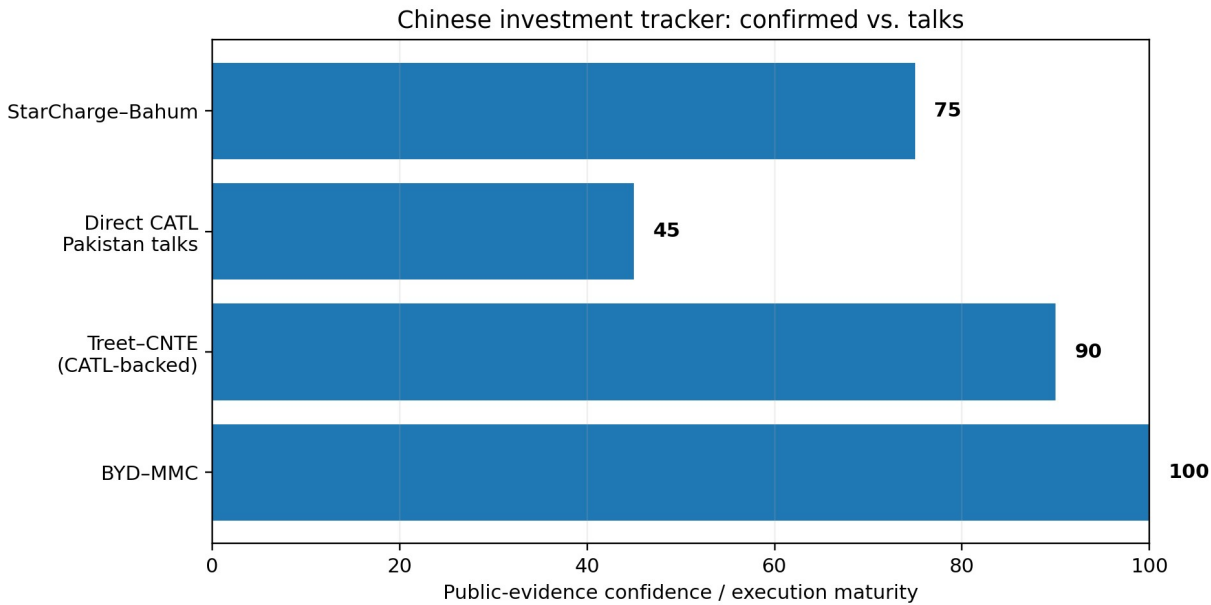


Chart 10. Public-evidence confidence/execution maturity. Scores are analytical and should be read with the deal-status table below.

Project	Status	Structure	Capacity / scope	Timeline	Classification
BYD–Mega Motor	Confirmed JV funded project	MMC 50:50 HPHL–Mega Conglomerate; BYD technology/supply arrangement	~USD150m; 25,000 vehicles/year CKD plant in Gharo	Commercial operations expected H2 2026	CONFIRMED
Treet–CNTE	Signed MoU	Treet + CNTE; CATL-backed CNTE	Energy-storage systems; CATL-cell-based products; future local assembly/value addition	MoU signed May 25, 2026	CONFIRMED MoU / NOT cell JV

Direct manufacturing	CATL	Talks / government engagement	Government–CATL discussions	Advanced batteries / storage; no public final plant capacity	Discussed May 2026	TALKS-STAGE
StarCharge–Bahum		Signed MoU	StarCharge Pakistan + Bahum Global	Charging ecosystem, local manufacturing and technology transfer	Announced 2026	July CONFIRMED MoU

CATL CLARIFICATION

CATL exposure in Pakistan should currently be assessed at two distinct levels. First, CNTE, a CATL-backed energy-storage company, has signed a formal MoU with Treet Battery, creating a credible pathway for CATL-linked technology, storage systems and potential local value addition. Second, direct discussions between the Government of Pakistan and CATL concerning advanced battery and storage investment remain at the engagement/talks stage. Accordingly, the Treet–CNTE agreement should be treated as a confirmed commercial relationship, while any direct CATL manufacturing commitment should remain classified as unconfirmed pending a definitive investment agreement, capacity announcement and implementation timetable.

6. Pakistan vs Bangladesh vs Vietnam — FDI competitiveness

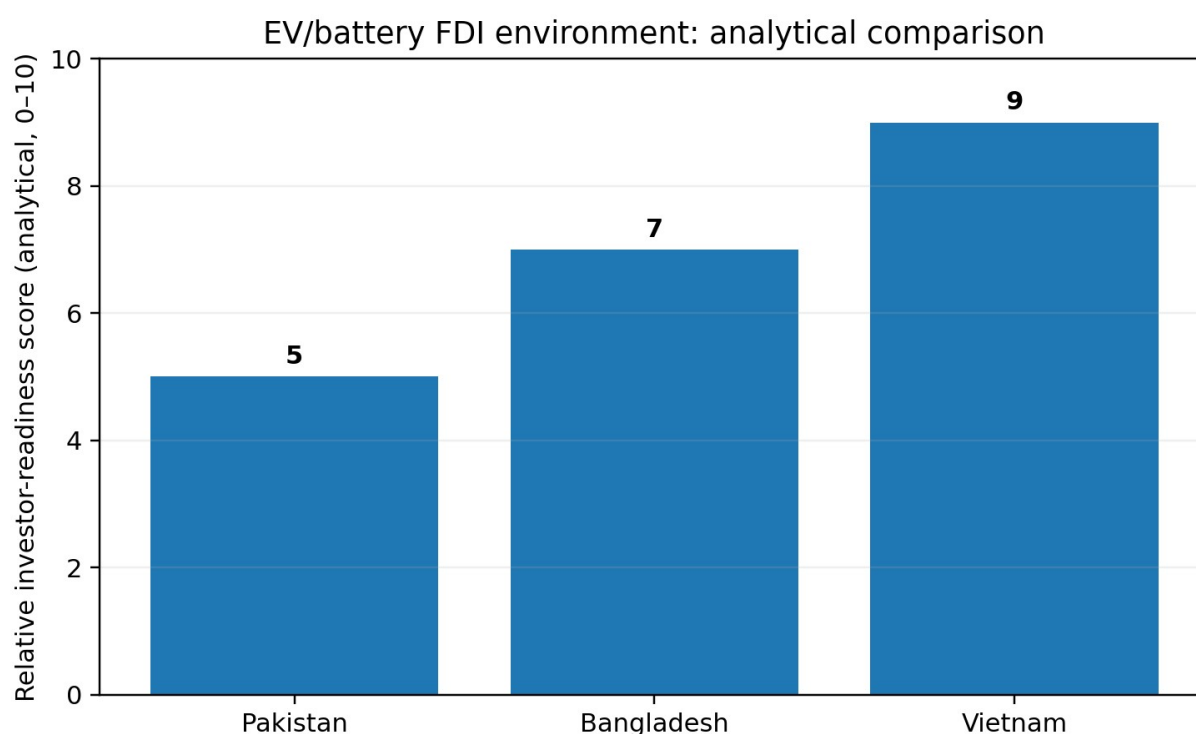


Chart 11. Analytical scorecard, not an official government ranking. Vietnam scores highest because it combines an established EV manufacturing ecosystem with broad investment incentives; Bangladesh is rapidly strengthening its EV manufacturing proposition; Pakistan has strong market-growth potential but greater policy and infrastructure uncertainty.

Dimension	Pakistan	Bangladesh	Vietnam	Investor takeaway
EV manufacturing incentives	Draft/transitioning; legacy EV CKD concessions	Proposed 2026 budget: low duties for higher value addition; support through 2031	Broad investment incentives + EV tax support	Pakistan needs policy certainty to compete
Battery/inputs	Policy direction; cells and processing gaps	Policy explicitly targets lithium cells/BESS localization	New-energy and supporting-industry investment incentives	Vietnam has deeper ecosystem; Bangladesh is building one

Tax / import support	Potentially strong but partly draft	Targeted exemptions and reduced duties	CIT, import-duty and land incentives available by qualifying project	Vietnam strongest for predictable project structuring
Market access	Large under-penetrated market	Large population and low car penetration	Large domestic EV manufacturing/export base	Pakistan's upside is high but demand data are less transparent
Infrastructure	Charging concentrated in major cities	Infrastructure developing	Much more mature EV ecosystem	Pakistan's charging gap is investable but a risk

6.1 What Bangladesh is changing

Bangladesh's 2026 budget proposals are notable because they explicitly tie lower duties to higher domestic value addition: high-value-added four-/three-wheeler manufacturing can receive very low import duty on inputs, while lower-value assembly receives a higher duty. Battery-related concessions are also scheduled to be withdrawn after June 2028 on the expectation that local production will have developed. That is a clearer 'build the ecosystem, then remove the crutch' design than an open-ended assembly subsidy.

6.2 Why Vietnam remains the benchmark

Vietnam combines a large-scale domestic EV manufacturing base with a formal investment-incentive architecture covering corporate income tax, import-duty exemptions for qualifying machinery/raw materials/components, land incentives and priority treatment for new energy and supporting industries. Battery/EV investors therefore enter an ecosystem rather than having to create the ecosystem from scratch.

7. 2026 policy timeline and monitoring triggers

Policy status timeline — 2026

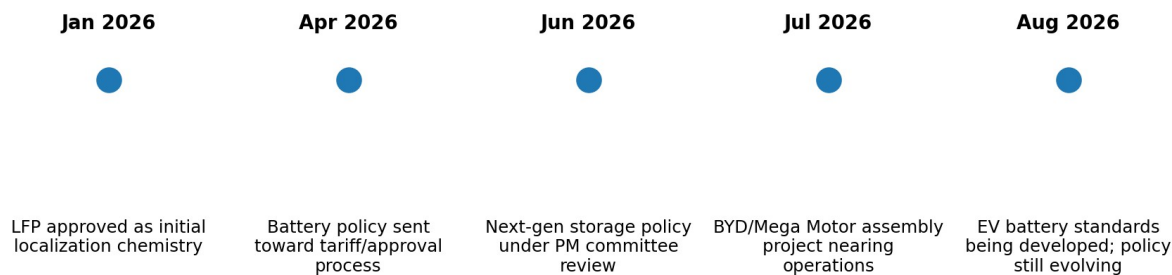


Chart 12. Key 2026 policy events used in this brief.

Trigger	What to monitor	Why it matters
Battery policy Gazette	Final localization tiers, tariff lines, sales-tax treatment, performance KPIs	Determines JV economics
PSQCA EV-battery standards	Cell/pack safety, certification, testing requirements	Can create mandatory capex and qualification barriers
Auto Policy 2026–31 notification	Local-content definition and tariff linkage	Determines vehicle assembly economics
Customs data by battery component	Cells vs modules vs packs vs cathodes/anodes	Best real-time localization test
Charging deployment outside big 3	Secondary cities + M2/M9/N5 corridors	Determines whether EV adoption can broaden
BYD Ghara line-off	Actual local content at launch	First major test of 'assembly vs manufacturing'
Treet–CNTE next-stage agreement	Whether MoU becomes local assembly/value-add	First CATL-linked industrialization test

8. Bottom line for investors

THE MARKET IS REAL; THE INDUSTRIAL DEPTH IS NOT THERE YET

Pakistan has the demand conditions for batteries: solar-led storage growth, rising EV/HEV interest, high petrol costs and a large import-dependent market. The country also has credible Chinese technology engagement. But the industrial stack remains incomplete. The most investable near-term opportunities are therefore not necessarily cell gigafactories: they include pack/module assembly, BMS, thermal systems, testing/certification, recycling, charging, and battery-grade mineral processing. Cell manufacturing is the strategic prize — but it requires policy certainty, quality standards, reliable power, foreign technical partners and a real upstream supply chain.

Recommended ARS tracking architecture: update monthly with five panels — (1) notified policy changes, (2) vehicle registrations by brand/powertrain, (3) battery import volumes by component, (4) charging

locations/capacity, and (5) JV status. Every data point should carry a source, date, geography and confidence grade.

Sources & methodology

Primary/official and high-quality sources used for this brief:

- Ministry of Industries & Production — Policies and EV/battery policy updates: <https://moip.gov.pk/Policies>
- Ministry of Industries & Production — May 2026 Chinese/Pakistani B2B meetings and EV Battery Policy status: <https://moip.gov.pk/NewsDetail/YTUwMWVIYzMTmYmYwNS00ODYxLTk2MDgtYWM1MTljYTlmYTgw>
- Associated Press of Pakistan — National Lithium-Ion Battery Manufacturing Policy 2026–31 review: <https://www.app.com.pk/business/sapm-haroon-chairs-meeting-to-review-lithium-ion-battery-policy/>
- Business Recorder — LFP localization and draft policy details: <https://www.brecorder.com/news/amp/40400951>
- Dawn — battery-policy status and EV battery standards, 18 Aug 2026: <https://www.dawn.com/news/2023581>
- Dawn — first lithium battery plant / Korangi project: <https://www.dawn.com/news/1990901>
- Pakistan Consulate General Guangzhou — Battery Energy Sector Investment Pitchbook: <https://www.pakcggz.gov.pk/trade-and-investments/investment-pitchbook/battery-energy-sector-investment-pitchbook-english-version>
- Renewables First / Energy Storage — 4.6 GWh BESS imports in 2025: <https://www.ess-news.com/2026/07/15/pakistan-imports-4-6-gwh-of-bess-in-2025/>
- Business Recorder — 2024 battery imports / 8.75 GWh 2030 projection: <https://www.brecorder.com/news/40366443/pakistans-lithium-battery-imports-soar-projected-to-hit-875-gwh-by-2030-report>
- Pakistan Automotive Manufacturers Association — monthly production and sales data: <https://pama.org.pk/monthly-production-sales-of-vehicles/>
- ProPakistani — March 2026 EV sales +61%: <https://propakistani.pk/2026/04/13/petrol-car-sales-decrease-by-9-and-electric-car-sales-increase-by-61-in-just-1-month-in-pakistan/>
- PakWheels — April 2026 model-wise sales: <https://www.pakwheels.com/blog/pakistan-car-sales-april-2026-model-wise-sales-breakdown/>
- Power Division — 44% discounted EV charging-station tariff / Rs39.70 per unit: <https://power.gov.pk/NewsDetail/YTczM2RiYTYtYTEyOC00YTBlWFmNzUtZWl2NTM5N2JhYmMy>
- NEPRA — EV charging tariff rationalization documents: <https://www.nepra.org.pk/Admission%20Notices/2025/02%20Feb/FEDERAL%20GOVERNMENT%20MOTION%20AND%20POLICY%20GUIDELINES%20FOR%20RATIONALIZATION%20OF%20TARIFF%20FOR%20ELECTRIC%20VEHICLE%20CHARGING%20STATIONS.PDF>
- Express Tribune — estimated 30–50 public charging stations and 3,000 target: <https://tribune.com.pk/story/2599316/evs-face-grid-land-trust-deficits>
- PACRA — Mega Motor Company rating report, JV structure and Gharo project: <https://www.pacra.com/api/rating-report/MTU4ODc%3D>
- HUBCO — Mega Motor Company: <https://hubpower.com/mega-motor/>
- Pakistan Stock Exchange disclosure via MarketScreener — Treet Battery/CNTE MoU: <https://www.marketscreener.com/news/treet-battery-signing-of-memorandum-of-understanding-ce7f5dd9d18ff727>
- Energy Storage — CATL-backed CNTE/Treet Battery agreement: <https://www.ess-news.com/2026/06/03/catl-backed-cnte-enters-pakistan-with-treet-battery-agreement/>
- Government of Pakistan — PM meeting with CATL and StarCharge: https://pid.gov.pk/site/press_detail/32856
- Pakistan Auto Industry Development and Export Policy 2021–26: <https://moip.gov.pk/SiteImage/Policy/Auto%20Industry%20Development%20and%20Export%20Policy%202021-26.pdf>
- PIDE — Automobile Policy 2026–31 recommendations and localization assessment: <https://pide.org.pk/research/automobile-policy-2026-2031-recommendations-from-pide-research/>
- Government of Bangladesh — FY2026–27 budget speech, EV manufacturing duty concessions: https://customs.gov.bd/files/BudgetSpeech_2026_2027_E.pdf

- Bangladesh Investment Development Authority — automobile incentives: <https://www.boi.gov.bd/automobile>
- Government of Vietnam — Investment Law 2025 and investment incentives: <https://en.baohinhphu.vn/viet-nams-law-on-investment-2025-111260731145214387.htm>
- Vietnam Investment — investment incentive sectors: <https://investvietnam.gov.vn/en/general-information-on-investment-regulations.nd/investment-incentives.html>
- USGS — Cambrian phosphate in Pakistan: https://pubs.usgs.gov/of/2015/1121/pdf/ofr20151121_report.pdf
- Geological Survey of Pakistan — lithium exploration in AJK / northern Pakistan: <https://gsp.gov.pk/geology-mineral-exploration/>
- Punjab Mines & Minerals — iron ore occurrences: https://mnmpunjab.gov.pk/important_minerals_occurring_in_punjab

ARS Data Brief • Pakistan EV & Battery Industrialization • 28 August 2026

Abbreviations

Abbreviation	Meaning
AIDEP	Auto Industry Development and Export Policy
ARS	Ardent Research Solutions
BESS	Battery Energy Storage System
BMS	Battery Management System
CBU	Completely Built Unit
CKD	Completely Knocked Down
CNTE	CATL-backed energy-storage company
CIT	Corporate Income Tax
EV	Electric Vehicle
EVCS	Electric Vehicle Charging Station
FDI	Foreign Direct Investment
FED	Federal Excise Duty
HEV	Hybrid Electric Vehicle
JV	Joint Venture
LFP	Lithium Iron Phosphate
MoU	Memorandum of Understanding
NEPRA	National Electric Power Regulatory Authority
NEV	New Energy Vehicle
OEM	Original Equipment Manufacturer
PAMA	Pakistan Automotive Manufacturers' Association
PHEV	Plug-in Hybrid Electric Vehicle
PSQCA	Pakistan Standards and Quality Control Authority
PKR / Rs	Pakistani Rupee
GWh	Gigawatt-hour
kWh	Kilowatt-hour
MW	Megawatt